

**A STUDY OF CONSENSUS MONTE CARLO:
METROPOLIS–HASTINGS ON DISTRIBUTED DATA**

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FINAL YEAR PROJECT REPORT

**A STUDY OF CONSENSUS MONTE CARLO:
METROPOLIS–HASTINGS ON DISTRIBUTED DATA**

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DECLARATION

I hereby declare that this thesis is based on my original work except for citations and quotations which have been duly acknowledged. I also declare that it has not been previously and concurrently submitted for any other degree or award at Xiamen University Malaysia or other institutions.

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APPROVAL FOR SUBMISSION

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ABSTRACT

In statistical inference, the frequentist approach treats parameters as unknown fixed constants, whereas the Bayesian framework views them as random variables following a probability distribution. While Monte Carlo methods allow for the computational approximation of these distributions, traditional centralized Markov Chain Monte Carlo (MCMC) becomes computationally prohibitive when dealing with large-scale or distributed datasets.

To address this, Consensus Monte Carlo offers a scalable alternative by partitioning data into shards, running separate MCMC chains locally, and combining the results to approximate the full posterior. This project implements a Consensus Monte Carlo framework using Independent Metropolis-Hastings (IMH) and Random Walk Metropolis-Hastings (RWMH). The study evaluates the accuracy of the consensus approach against a full-data Metropolis-Hastings baseline to determine the conditions under which the distributed method is an adequate substitute.

Keywords: Bayesian Inference, Markov Chain Monte Carlo (MCMC), Metropolis-Hastings, Consensus Monte Carlo.

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LIST OF SYMBOLS / ABBREVIATIONS

List of Symbols/Abbreviations is optional. Delete this chapter if you do not want to include it. Delete this red paragraph if you want to include it, and remember to modify the list to include only those symbols/abbreviations that appear in your own thesis.

- $\mathbf{r}(t)$: Parametric curve
- ∇ : Gradient operator
- $\langle, \rangle$: Inner product
- CLT : Central limit theorem

CHAPTER 1

INTRODUCTION

1.1 Background Study

Bayesian inference is a statistical framework for learning about unknown quantities through probability. Let θ denote a vector of unknown model parameters and let y denote the observed data. In the Bayesian approach, prior information about θ is represented by a prior distribution $\pi(\theta)$, while the information provided by the observed data is represented by the likelihood function $L(y | \theta)$. These two components are combined through Bayes' theorem to obtain the posterior distribution,

$$\pi(\theta | y) = \frac{L(y | \theta)\pi(\theta)}{p(y)},$$

where $p(y)$ is the marginal likelihood. Since $p(y)$ does not depend on θ , the posterior distribution is commonly written as

$$\pi(\theta | y) \propto L(y | \theta)\pi(\theta).$$

The posterior distribution $\pi(\theta | y)$ is central in Bayesian inference because it describes the uncertainty about the unknown parameters after the data have been observed. Posterior means, posterior medians, posterior standard deviations, and credible intervals can be computed from this distribution when posterior samples or analytic expressions are available.

In many practical models, the posterior distribution cannot be evaluated directly in closed form. This difficulty often arises because the marginal likelihood involves integration over the parameter space, and the resulting posterior summaries may also require high-dimensional integration. Markov Chain Monte Carlo (MCMC) methods provide a computational approach for approximating posterior distributions by generating dependent samples from a Markov chain whose stationary distribution is the target posterior distribution. TODO: add citation for Bayesian inference and MCMC.

The Metropolis–Hastings algorithm is a general MCMC method for sampling from a target distribution that is known up to a proportionality constant. Starting from a current parameter value, the algorithm proposes a candidate value from a proposal distribution and then

accepts or rejects the candidate according to an acceptance probability determined by the target posterior density and the proposal mechanism. When the Markov chain is run for a sufficiently long period under suitable regularity conditions, the retained samples can be used to approximate posterior summaries. TODO: add citation for the Metropolis–Hastings algorithm.

For large datasets, full-data MCMC can become computationally expensive. If the data contain n observations, then each iteration of a full-data Metropolis–Hastings sampler may require evaluating the likelihood contribution of all n observations. As n increases, the cost of producing a sufficiently long Markov chain can become substantial. This issue is especially important when repeated likelihood evaluations are required for many proposed parameter values.

Distributed data settings provide one possible way to reduce this computational burden. Instead of treating the dataset as a single object processed by one sampler, the observations may be divided into K subsets. Let y_m denote the data in subset m , where $m = 1, \dots, K$. Each subset can be processed separately, which allows the computational work to be divided across independent chains or computing resources.

Consensus Monte Carlo is a distributed Bayesian computation method based on this idea. The full posterior is approximated by first constructing subposteriors from the data subsets, running MCMC separately on each subset, and then combining the resulting subposterior samples to form an approximation to the full posterior distribution [Sco17, SBB⁺22]. Suppose the full dataset is partitioned into K subsets, $y_1, \dots, y_K$, and the likelihood can be factorised as

$$L(y \mid \theta) = \prod_{m=1}^K L(y_m \mid \theta).$$

In Consensus Monte Carlo, a common construction defines the m th subposterior as

$$\pi_m(\theta \mid y_m) \propto L(y_m \mid \theta)\pi(\theta)^{1/K},$$

so that the product of the subposteriors satisfies

$$\prod_{m=1}^K \pi_m(\theta \mid y_m) \propto \prod_{m=1}^K L(y_m \mid \theta)\pi(\theta)^{1/K} = L(y \mid \theta)\pi(\theta) \propto \pi(\theta \mid y).$$

Thus, the product of the subposteriors is proportional to the full posterior under this factorisation. In this project, Consensus Monte Carlo is studied together with the Metropolis–Hastings algorithm for distributed data. The current application uses UK accident data, and the detailed implementation is reported in the methodology and results chapters.

1.2 Problem Statement

Bayesian inference provides a coherent method for estimating unknown model parameters and quantifying uncertainty through the posterior distribution. However, its practical implementation can be computationally demanding when posterior simulation is required for a large dataset. In a full-data Metropolis–Hastings algorithm, the likelihood is commonly evaluated using all observations at each iteration. Therefore, the computational cost may increase considerably as the dataset becomes larger.

Distributed computation can reduce this burden by separating the full dataset into smaller subsets and allowing the subset analyses to be carried out independently. Consensus Monte Carlo uses this principle by running MCMC on each subset and combining the resulting subposterior samples to approximate the full posterior distribution. This approach is attractive because the subset chains can be run separately and may require less computation per iteration than a full-data chain.

The main statistical issue is whether the consensus approximation is sufficiently close to the posterior distribution obtained from full-data Metropolis–Hastings sampling. The approximation may be affected by the number of subsets, the way the data are partitioned, the prior adjustment used for the subposteriors, the behaviour of the subset Markov chains, and the posterior combination method. Consequently, it is necessary to compare the consensus posterior with the full-data posterior rather than assuming that the distributed method automatically gives equivalent results.

This project studies Consensus Monte Carlo using Metropolis–Hastings on distributed data. The comparison focuses on posterior summaries, credible intervals, convergence behaviour, and computational efficiency. The detailed empirical findings are reported in the results chapter, and no claim about accuracy, runtime advantage, or practical superiority is made without support from the completed analysis.

1.3 Research Objectives

The objectives of this project are as follows:

1. To study Bayesian inference and Metropolis–Hastings algorithms for posterior estimation.
2. To implement full-data Metropolis–Hastings sampling.
3. To implement Consensus Monte Carlo by dividing the dataset into subsets and running Metropolis–Hastings separately on each subset.
4. To combine subposterior samples to approximate the full posterior distribution.
5. To compare full-data Metropolis–Hastings and Consensus Monte Carlo in terms of posterior summaries, credible intervals, convergence behaviour, and computational efficiency.

1.4 Research Framework

The research framework begins with data preparation. The dataset is inspected, cleaned, and arranged into a form suitable for Bayesian modelling. This stage may include identifying the response variable, selecting possible predictor variables, handling missing or invalid observations, and constructing the design matrix X . The selected variables and preprocessing decisions are reported in the methodology and results chapters.

The second stage is model specification. A Bayesian model is defined by specifying the response variable y , the predictor matrix X , the parameter vector θ , the likelihood function $L(y \mid \theta)$, and the prior distribution $\pi(\theta)$. The resulting target distribution for the full-data analysis is the posterior distribution $\pi(\theta \mid y)$.

The third stage is full-data Metropolis–Hastings sampling. In this stage, a Markov chain is constructed to sample from the posterior distribution based on the complete dataset. The resulting samples provide the baseline posterior summaries for comparison with the distributed approach. Algorithm settings such as proposal distributions, number of iterations, burn-in, thinning, and random seed are reported in the methodology and results chapters.

The fourth stage is data partitioning. The complete dataset is divided into K subsets, denoted by $y_1, \dots, y_K$. Each subset is used to define a subposterior distribution. The value of K and the partitioning strategy will be determined and documented in the methodology and results chapters.

The fifth stage is subset Metropolis–Hastings sampling. Separate Metropolis–Hastings chains are run on each subset to obtain subposterior samples. If θ_m denotes a subset-specific parameter draw associated with subset m , then the collection of samples from all subsets is used as the input for the consensus step.

The sixth stage is consensus posterior combination. The subposterior samples are combined to construct an approximation to the full posterior distribution. The combination method will be described in detail in the methodology chapter, and its empirical performance will be evaluated in the results chapter.

The final stage is comparison and conclusion. The full-data posterior and the consensus posterior are compared using posterior means, posterior medians, posterior standard deviations, credible intervals, convergence behaviour, and computational efficiency where these quantities are available from the completed analysis. Any unavailable diagnostics or numerical summaries will be marked clearly as TODO items until they are computed from the final R script.

CHAPTER 2

LITERATURE REVIEW

2.1 Comparing Consensus Monte Carlo Strategies for Distributed Bayesian Computation

This paper by Steven L. Scott reviews and compares methods for distributed Bayesian computation, focusing on how to combine posterior samples generated separately on different data partitions. Its main concern is the consensus Monte Carlo setting, where large datasets are split across machines to reduce computational burden and communication costs. Scott evaluates three main aggregation strategies already discussed in the literature: averaging methods, sequential resampling approaches, and density-estimation methods. The paper shows that simple weighted averaging performs very well when subposteriors are close to Gaussian, which often happens in large-sample settings, making it a strong practical baseline despite its simplicity.

A major contribution of the paper is its critical comparison of these methods and the proposal of a mixture-based consensus Monte Carlo approach, which uses Gaussian mixtures to better capture non-Gaussian posterior shapes. Scott finds that while more flexible methods such as kernel-based and sequential approaches are attractive in theory, they tend to perform poorly as dimension increases, whereas averaging remains stable and computationally efficient. The paper therefore concludes that in distributed Bayesian inference, the main challenge is not generating local posterior samples but combining them accurately without losing scalability. Overall, the study is important because it gives a balanced assessment of the trade-off between statistical accuracy and computational practicality in scalable Bayesian analysis. [Sco17]

2.2 Bayes and Big Data: The Consensus Monte Carlo Algorithm

Scott et al.'s Bayes and Big Data: The Consensus Monte Carlo Algorithm sits in the early literature on scalable Bayesian computation, addressing the problem of inference when data are too large for a single machine. The paper argues that existing parallel approaches based

on multicore CPUs or GPUs reduce processor bottlenecks but do not solve memory and disk limits, while multi-machine settings introduce severe communication costs that make standard synchronized MCMC inefficient. In positioning their method, the authors connect their work to several adjacent strands: averaging frequentist estimators in distributed settings, the “bag of little bootstraps,” earlier Bayesian consensus ideas by Huang and Gelman, and deterministic Bayesian approximations such as INLA and variational Bayes. Their literature framing is therefore important because it shows consensus Monte Carlo not as an isolated algorithm, but as one response within a broader effort to make inference feasible under big-data constraints while still preserving access to the full posterior distribution.

The paper’s main contribution is to propose consensus Monte Carlo, where data are split into shards, each worker runs an independent Monte Carlo simulation on its local subproblem, and the resulting draws are combined through averaging rules to approximate the full posterior. In the examples, the method performs well not only in Gaussian settings but also in some non-Gaussian cases, including beta-binomial models, hierarchical models, and BART, which supports the authors’ claim that simple averaging can be surprisingly effective in distributed Bayesian analysis. At the same time, the paper is careful about limitations: the method is best suited to fixed-dimensional continuous parameter spaces, and the authors explicitly note that performance away from Gaussianity remains an open question, with more sophisticated consensus mechanisms still needing development. As a result, this paper is best read as a foundational contribution that establishes the practicality of low-communication Bayesian aggregation, while also defining the main theoretical and methodological issues that later distributed MCMC research would need to resolve. [SBB⁺22]

2.3 Sampling for Bayesian computation with large datasets

Huang and Gelman’s Sampling for Bayesian Computation with Large Datasets is an early contribution to the literature on scalable Bayesian inference for hierarchical models. The paper begins from the observation that Bayesian multilevel modeling had already become important across fields such as health, education, environment, and business, but that standard Gibbs and related MCMC methods became difficult to use once datasets were too large for convenient storage and repeated full-data iteration. In that context, the authors position their work against earlier data-subsetting approaches, especially sequential and

importance-sampling based methods such as Chopin (2002) and Ridgeway and Madigan (2002), arguing that their own strategy extends this line of work by using random partitioning and cluster sampling so that each subset preserves the hierarchical structure of the original data. This framing is important because it treats big-data Bayes not as a completely new problem, but as an extension of existing work on approximate posterior computation under computational constraints.

The paper's main methodological contribution is a consensus-style posterior combination strategy: divide the data into disjoint subsets, fit the Bayesian model separately on each subset, and then combine the resulting inferences using a normal approximation, with possible refinement through importance sampling. In the literature, this paper is therefore best understood as a precursor to later consensus Monte Carlo methods, because it already identifies the central ideas that would define that area: subset-based inference, low communication cost, and the statistical problem of how to recombine local posterior information into a global approximation. The authors are also careful to state the unresolved issues, including the lack of a full theoretical justification, the choice of the number of subsets K , and the question of which aggregation rule is best. As a result, the paper is significant less for providing a final solution than for establishing the basic research agenda for distributed Bayesian computation in large hierarchical datasets. [HG05]

CHAPTER 3

METHODOLOGY

3.1 Data Source

The empirical application in this project uses UK accident data. TODO: add citation for the UK accident dataset source. The data contain accident-level information, including accident severity and road-related variables. For reproducibility, the data file used in the analysis is stored as `Dataset/accident_2013.csv`, and the accompanying variable documentation is provided in `Dataset/Variables for UK data.pdf`. The detailed description of the variables used in the empirical analysis is reported together with the data preparation and results.

The purpose of using this dataset is to provide an applied setting for comparing full-data Metropolis–Hastings sampling with Consensus Monte Carlo. The analysis focuses on a binary response derived from accident severity. In this study, slight accidents are modelled against the other severity categories.

3.2 Data Preprocessing

The data preprocessing stage prepares the raw accident data for Bayesian logistic regression. The empirical analysis uses accident severity, road type, and speed limit from the dataset. Road type and speed limit are treated as categorical predictors, and observations with missing values in the selected variables are removed before model fitting.

The original accident severity variable contains more than two categories. For the logistic regression model, it is transformed into a binary response variable. Let y_i denote the response for observation i . In this study,

$$y_i = \begin{cases} 1, & \text{if observation } i \text{ corresponds to a slight accident,} \\ 0, & \text{otherwise.} \end{cases}$$

Thus, the event being modelled is the occurrence of a slight accident relative to the other accident severity categories. The categorical predictors are converted into a design matrix X using indicator variables, with the intercept included in the model matrix.

The empirical analysis also removes observations with unknown road type before fitting the logistic regression model. The exact preprocessing decisions and the resulting descriptive summaries are reported in the results chapter. No numerical sample sizes or category counts are stated here unless they are obtained from the analysis output.

3.3 Bayesian Logistic Regression Model

Let $y_i \in \{0, 1\}$ be the binary accident severity response for observation i , and let x_i be the corresponding predictor vector, including the intercept term and the encoded categorical predictors. Let $\beta = (\beta_1, \dots, \beta_d)^T$ denote the vector of regression coefficients, where d is the number of regression coefficients in the model. The Bayesian logistic regression model assumes that

$$y_i \mid \beta \sim \text{Bernoulli}(p_i),$$

where p_i is the probability that observation i is classified as a slight accident. The logistic link function is used:

$$\text{logit}(p_i) = x_i^T \beta.$$

Equivalently,

$$p_i = \frac{\exp(x_i^T \beta)}{1 + \exp(x_i^T \beta)}.$$

For n independent observations, the likelihood function is

$$L(y \mid \beta, X) = \prod_{i=1}^n p_i^{y_i} (1 - p_i)^{1-y_i}.$$

In computation, the log-likelihood is used for numerical stability:

$$\log L(y \mid \beta, X) = \sum_{i=1}^n [y_i x_i^T \beta - \log\{1 + \exp(x_i^T \beta)\}].$$

The computation evaluates this expression using a numerically stable form of $\log\{1 + \exp(x_i^T \beta)\}$. TODO: add citation for Bayesian logistic regression.

3.4 Prior Distribution

A normal prior distribution is assigned independently to each regression coefficient. For coefficient β_j ,

$$\beta_j \sim N(0, \sigma_\beta^2),$$

where σ_β is the prior standard deviation. In this study, the prior standard deviation is set to $\sigma_\beta = 10$. This prior is centred at zero and gives a relatively diffuse prior distribution for the regression coefficients in the logistic regression model.

Under independence of the prior components, the prior density for β can be written as

$$\pi(\beta) = \prod_{j=1}^d \pi(\beta_j),$$

where d is the number of regression coefficients.

3.5 Posterior Distribution

Combining the likelihood and the prior distribution gives the posterior distribution of the regression coefficient vector:

$$\pi(\beta \mid y, X) \propto L(y \mid \beta, X)\pi(\beta).$$

The posterior distribution is not summarised analytically in this project. Instead, posterior samples are generated using Metropolis–Hastings algorithms. TODO: add citation for posterior simulation.

3.6 Full-Data Metropolis–Hastings Sampling

The full-data analysis uses the complete response vector y and design matrix X to sample from $\pi(\beta \mid y, X)$. Two Metropolis–Hastings approaches are considered: Random Walk Metropolis–Hastings and Independent Metropolis–Hastings. TODO: add citation for the Metropolis–Hastings algorithm. In both cases, the Markov chain is initialised using estimates from a frequentist logistic regression fit, and non-finite starting values are replaced by zero in the computation.

The full-data sampler provides a baseline approximation to the posterior distribution. The posterior summaries from this baseline are later compared with the posterior summaries obtained from Consensus Monte Carlo. Algorithm settings such as the number of iterations, burn-in, tuning constants, and proposal covariance choices are implementation details reported with the computational results.

3.6.1 Random Walk Metropolis–Hastings Algorithm

In Random Walk Metropolis–Hastings, the proposal at iteration t is generated by perturbing the previous state. If $\beta^{(t-1)}$ is the current value, then the proposed value is

$$\beta^* = \beta^{(t-1)} + \epsilon,$$

where

$$\epsilon \sim N(0, \Sigma_{\text{RWMH}}).$$

Here, Σ_{RWMH} is the proposal covariance matrix. In this study, the random walk sampler uses a multivariate normal proposal with covariance based on the covariance matrix from the fitted logistic regression model and a tuning factor.

Since the random walk proposal is symmetric, the proposal densities cancel in the Metropolis–Hastings ratio. The acceptance probability is therefore

$$\alpha = \min \left\{ 1, \frac{\pi(\beta^* | y, X)}{\pi(\beta^{(t-1)} | y, X)} \right\}.$$

In log form, the proposal is accepted when

$$\log u < \log \pi(\beta^* | y, X) - \log \pi(\beta^{(t-1)} | y, X),$$

where $u \sim \text{Uniform}(0, 1)$.

3.6.2 Independent Metropolis–Hastings Algorithm

In Independent Metropolis–Hastings, the proposed value is generated from a fixed proposal distribution rather than by adding a random perturbation to the current state. The empirical analysis uses a multivariate normal proposal distribution:

$$\beta^* \sim q(\beta),$$

where $q(\beta)$ is a fixed multivariate normal density. The proposal mean is based on the frequentist logistic regression estimate, and the proposal covariance is based on the corresponding estimated covariance matrix with a tuning factor.

Because the proposal distribution is not a symmetric random walk, the proposal-density correction must be included in the acceptance probability. If the current value is $\beta^{(t-1)}$, then

$$\alpha = \min \left\{ 1, \frac{\pi(\beta^* | y, X)q(\beta^{(t-1)})}{\pi(\beta^{(t-1)} | y, X)q(\beta^*)} \right\}.$$

Equivalently, the log acceptance ratio is

$$\log \pi(\beta^* \mid y, X) - \log \pi(\beta^{(t-1)} \mid y, X) + \log q(\beta^{(t-1)}) - \log q(\beta^*).$$

3.7 Consensus Monte Carlo

Consensus Monte Carlo is used to approximate the full posterior distribution by dividing the data into subsets, sampling from subset-specific subposteriors, and combining the subposterior samples. TODO: add citation for Consensus Monte Carlo. This method is considered in order to study whether a distributed approximation can produce posterior summaries close to those from the full-data Metropolis–Hastings analysis.

3.7.1 Data Partitioning

Let the full dataset be partitioned into K subsets, where K denotes the number of data subsets. The response vector and design matrix for subset m are denoted by y_m and X_m , respectively, where $m = 1, \dots, K$. In the empirical analysis, observations are randomly assigned to subsets before the subset samplers are run. The value of K is treated as an implementation setting and is reported with the computational results.

The full likelihood can be factorised across the subsets as

$$L(y \mid \beta, X) = \prod_{m=1}^K L(y_m \mid \beta, X_m).$$

3.7.2 Subposterior Distribution

For subset m , the subposterior distribution is defined by combining the subset likelihood with a fraction of the prior:

$$\pi_m(\beta \mid y_m, X_m) \propto L(y_m \mid \beta, X_m) \pi(\beta)^{1/K}.$$

The factor $\pi(\beta)^{1/K}$ distributes the prior contribution across the K subsets. This construction gives

$$\prod_{m=1}^K \pi_m(\beta \mid y_m, X_m) \propto \prod_{m=1}^K L(y_m \mid \beta, X_m) \pi(\beta)^{1/K} = L(y \mid \beta, X) \pi(\beta) \propto \pi(\beta \mid y, X).$$

Therefore, under the likelihood factorisation, the product of the subposterior densities is proportional to the full posterior density.

3.7.3 Subset Metropolis–Hastings Sampling

Separate Metropolis–Hastings chains are run on each subset to obtain samples from the subposterior distributions. The empirical analysis applies both subset Random Walk Metropolis–Hastings and subset Independent Metropolis–Hastings. For each subset, the target density in the acceptance probability is the subposterior density $\pi_m(\beta \mid y_m, X_m)$ rather than the full posterior density.

For subset Random Walk Metropolis–Hastings, the proposal is centred at the current value and uses a multivariate normal random walk proposal. For subset Independent Metropolis–Hastings, the proposal is generated from a fixed multivariate normal proposal distribution. In the independent proposal case, the proposal-density correction is included in the acceptance probability.

After burn-in is removed, the retained subposterior samples from each subset are used in the consensus combination step. The number of retained samples is aligned across subsets by using the same available sample length from each subset chain.

3.7.4 Consensus Posterior Combination

The subposterior samples are combined using a precision-weighted consensus method. Let S denote the common number of retained posterior samples after burn-in, and let $\beta_m^{(s)}$ denote the s th retained sample from subset m , where $s = 1, \dots, S$. Let W_m denote the estimated precision matrix for subset m , obtained as the inverse of the estimated covariance matrix of the subposterior samples from that subset. The consensus sample at index s is defined as

$$\beta_{\text{consensus}}^{(s)} = \left(\sum_{m=1}^K W_m \right)^{-1} \sum_{m=1}^K W_m \beta_m^{(s)}.$$

This produces a sequence of consensus posterior samples

$$\left\{ \beta_{\text{consensus}}^{(s)} \right\}_{s=1}^S,$$

where S is the common number of retained samples used from the subset chains after burn-in. These samples are then summarised and compared with the full-data posterior samples.

3.8 Posterior Summary and Evaluation Criteria

The posterior samples obtained from full-data Metropolis–Hastings and Consensus Monte Carlo are evaluated using posterior summaries and diagnostic quantities. For each regression coefficient, the following summaries are considered:

1. posterior mean;
2. posterior median;
3. posterior standard deviation;
4. 95% credible interval;
5. effective sample size;
6. Monte Carlo standard error.

The empirical analysis estimates effective sample size using the autocorrelation structure of the sampled chain and computes the Monte Carlo standard error as the posterior standard deviation divided by the square root of the estimated effective sample size.

Additional evaluation criteria include acceptance rate, trace plots, and posterior histograms or density plots. The acceptance rate is used to assess the behaviour of the Metropolis–Hastings proposal mechanism. Trace plots are used to visually inspect the mixing behaviour and stability of the chains. Posterior histograms or density plots are used to compare the marginal posterior distributions produced by the full-data and consensus approaches.

Computational efficiency is also considered if runtime information is available from the analysis output. In the results chapter, the comparison between full-data Metropolis–Hastings and Consensus Monte Carlo is based on the posterior summaries, credible intervals, diagnostic measures, graphical checks, and available computational timing results. No numerical conclusion is stated unless it is supported by the generated output.

CHAPTER 4

RESULTS AND ANALYSIS

4.1 Data Summary

This chapter reports the empirical results from the Bayesian logistic regression analysis. The empirical analysis uses UK accident data with a binary accident severity response, where slight accidents are modelled against the other severity categories. The model uses Bayesian logistic regression with the predictors and preprocessing steps described in Chapter 3.

Table 4.1 summarises the computational settings used for the reported analysis. After preprocessing, the analysis contains 82,344 observations and 10 regression coefficients. The full-data samplers use 100,000 MCMC iterations, while the subset samplers for Consensus Monte Carlo use 50,000 MCMC iterations for each subset. The burn-in fraction is 0.2, and the Consensus Monte Carlo implementation uses $K = 4$ subsets.

Table 4.1: Model settings for the empirical analysis

Setting	Value
Observations after preprocessing	82344
Number of coefficients	10
Full-data MCMC iterations	100000
Burn-in fraction	0.2
Number of subsets, K	4
Subset MCMC iterations	50000

The computational output does not contain a descriptive statistics table for the original variables or response distribution. Therefore, descriptive summaries by accident severity, road type, and speed limit are left for later inclusion.

TODO: add descriptive statistics table for the response and predictors when the output is available.

4.2 Full-Data Metropolis–Hastings Results

The full-data analysis was performed using Random Walk Metropolis–Hastings and Independent Metropolis–Hastings. The posterior summaries available for both full-data samplers include posterior mean, posterior median, posterior standard deviation, effective sample size, Monte Carlo standard error, 95% credible interval, and odds-ratio summaries.

The regression coefficients are interpreted relative to the reference categories used in the design matrix. The exact reference categories for road type and speed limit should be stated once the design matrix coding is documented. TODO: identify the exact reference category for road type. TODO: identify the exact reference category for speed limit.

For logistic regression, exponentiating a coefficient gives the corresponding odds ratio. An odds ratio greater than one indicates higher posterior estimated odds of a slight accident relative to the relevant reference category or baseline condition, whereas an odds ratio less than one indicates lower posterior estimated odds. The odds-ratio summaries are available in the computational output, but separate odds-ratio tables are not added here to keep the main results tables compact.

4.2.1 Random Walk Metropolis–Hastings

The full-data Random Walk Metropolis–Hastings sampler has an acceptance rate of 0.39195. Table 4.2 reports selected posterior summaries for the regression coefficients. The posterior summaries are reported on the log-odds scale, while odds-ratio summaries are available in the computational output.

Table 4.2: Full-data Random Walk Metropolis–Hastings posterior summaries

Parameter	Mean	Median	SD	Lower 95% CrI	Upper 95% CrI
Intercept	1.9517	1.9507	0.0812	1.7955	2.1140
Road type: one-way street	-0.2126	-0.2129	0.0817	-0.3719	-0.0515
Road type: roundabout	0.1855	0.1856	0.0493	0.0900	0.2816
Road type: single carriageway	-0.2267	-0.2263	0.0390	-0.3038	-0.1517
Road type: slip road	0.5335	0.5306	0.1195	0.3062	0.7768
Speed limit 2	0.1730	0.1742	0.0717	0.0296	0.3082
Speed limit 3	-0.0089	-0.0088	0.0794	-0.1645	0.1434
Speed limit 4	-0.2127	-0.2124	0.0908	-0.3922	-0.0370
Speed limit 5	-0.3658	-0.3648	0.0765	-0.5195	-0.2191
Speed limit 6	-0.2205	-0.2193	0.0985	-0.4144	-0.0310

For this sampler, the posterior means and posterior medians are close for all reported coefficients. The 95% credible interval for speed limit 3 includes zero, while the remaining reported intervals do not include zero.

4.2.2 Independent Metropolis–Hastings

The full-data Independent Metropolis–Hastings sampler has an acceptance rate of 0.21504. Table 4.3 reports selected posterior summaries. The posterior means from the two full-data samplers are very close, suggesting that the two full-data Metropolis–Hastings implementations produce similar posterior summaries for this output.

Table 4.3: Full-data Independent Metropolis–Hastings posterior summaries

Parameter	Mean	Median	SD	Lower 95% CrI	Upper 95% CrI
Intercept	1.9500	1.9493	0.0797	1.7941	2.1116
Road type: one-way street	-0.2130	-0.2142	0.0813	-0.3701	-0.0539
Road type: roundabout	0.1867	0.1859	0.0486	0.0903	0.2824
Road type: single carriageway	-0.2246	-0.2248	0.0388	-0.3014	-0.1509
Road type: slip road	0.5370	0.5367	0.1187	0.3090	0.7746
Speed limit 2	0.1729	0.1751	0.0713	0.0297	0.3101
Speed limit 3	-0.0091	-0.0074	0.0789	-0.1672	0.1425
Speed limit 4	-0.2108	-0.2102	0.0904	-0.3887	-0.0360
Speed limit 5	-0.3652	-0.3640	0.0769	-0.5177	-0.2172
Speed limit 6	-0.2189	-0.2182	0.0987	-0.4152	-0.0245

4.3 Consensus Monte Carlo Results

Consensus Monte Carlo was implemented using $K = 4$ subsets. The computational output contains posterior summaries for Consensus Monte Carlo using Random Walk Metropolis–Hastings and Independent Metropolis–Hastings. The subset acceptance rates are reported in Table 4.4.

4.3.1 Consensus Monte Carlo using Random Walk Metropolis–Hastings

Table 4.5 reports the posterior summaries for Consensus Monte Carlo using Random Walk Metropolis–Hastings. The CMC-RWMH posterior summaries are generally close to the full-data posterior summaries, although small differences are present for some coefficients.

The CMC-RWMH subset acceptance rates range from 0.38446 to 0.40850. These values are close to the full-data RWMH acceptance rate of 0.39195.

Table 4.4: Acceptance rates from the computational output

Method	Acceptance rate
Full RWMH	0.39195
Full IMH	0.21504
CMC-RWMH subset 1	0.40850
CMC-RWMH subset 2	0.39748
CMC-RWMH subset 3	0.39106
CMC-RWMH subset 4	0.38446
CMC-IMH subset 1	0.02318
CMC-IMH subset 2	0.14236
CMC-IMH subset 3	0.03542
CMC-IMH subset 4	0.01420

Table 4.5: Consensus Monte Carlo using Random Walk Metropolis–Hastings posterior summaries

Parameter	Mean	Median	SD	Lower 95% CrI	Upper 95% CrI
Intercept	1.9485	1.9473	0.0807	1.7914	2.1085
Road type: one-way street	-0.2128	-0.2132	0.0824	-0.3742	-0.0522
Road type: roundabout	0.1790	0.1786	0.0497	0.0812	0.2757
Road type: single carriageway	-0.2334	-0.2334	0.0389	-0.3092	-0.1569
Road type: slip road	0.5123	0.5111	0.1219	0.2786	0.7513
Speed limit 2	0.1830	0.1832	0.0726	0.0409	0.3250
Speed limit 3	0.0010	0.0008	0.0799	-0.1548	0.1576
Speed limit 4	-0.2131	-0.2130	0.0918	-0.3953	-0.0300
Speed limit 5	-0.3590	-0.3589	0.0780	-0.5130	-0.2076
Speed limit 6	-0.2222	-0.2241	0.0987	-0.4152	-0.0275

4.3.2 Consensus Monte Carlo using Independent Metropolis–Hastings

The computational output contains posterior summaries for Consensus Monte Carlo using Independent Metropolis–Hastings. Therefore, the CMC-IMH results are reported in Table 4.6. The subset acceptance rates for CMC-IMH are lower and more variable than those for CMC-RWMH, with values ranging from 0.01420 to 0.14236. These low subset acceptance rates suggest that the CMC-IMH sampler may require further tuning before strong conclusions are drawn from this method.

Table 4.6: Consensus Monte Carlo using Independent Metropolis–Hastings posterior summaries

Parameter	Mean	Median	SD	Lower 95% CrI	Upper 95% CrI
Intercept	1.9546	1.9575	0.0782	1.7897	2.1005
Road type: one-way street	-0.1850	-0.1838	0.0801	-0.3413	-0.0300
Road type: roundabout	0.1833	0.1832	0.0486	0.0871	0.2786
Road type: single carriageway	-0.2300	-0.2305	0.0387	-0.3042	-0.1542
Road type: slip road	0.5223	0.5200	0.1145	0.3000	0.7563
Speed limit 2	0.1696	0.1687	0.0687	0.0376	0.3070
Speed limit 3	-0.0043	-0.0062	0.0792	-0.1587	0.1590
Speed limit 4	-0.2159	-0.2191	0.0900	-0.3886	-0.0303
Speed limit 5	-0.3534	-0.3553	0.0757	-0.4969	-0.2040
Speed limit 6	-0.2260	-0.2265	0.0919	-0.4070	-0.0422

4.4 Comparison of Posterior Estimates

Table 4.7 compares posterior means across the four methods. The posterior means from the two full-data samplers are very close. The CMC-RWMH posterior means are also generally close to the full-data posterior means. The CMC-IMH posterior means are broadly similar for most parameters, although the result for the one-way street coefficient differs more visibly from the full-data estimates than the corresponding CMC-RWMH estimate.

The credible intervals are broadly similar across methods for most parameters. For all four methods, the 95% credible interval for speed limit 3 includes zero. For the remaining reported coefficients, the available 95% credible intervals do not include zero. This statement is descriptive and should not be interpreted as an assessment of practical importance without further model checking and contextual interpretation.

The effective sample size and Monte Carlo standard error are available in the posterior summaries. The full-data IMH output has larger effective sample sizes than the full-data

Table 4.7: Comparison of posterior means across methods

Parameter	Full RWMH	Full IMH	CMC-RWMH	CMC-IMH
Intercept	1.9517	1.9500	1.9485	1.9546
Road type: one-way street	-0.2126	-0.2130	-0.2128	-0.1850
Road type: roundabout	0.1855	0.1867	0.1790	0.1833
Road type: single carriageway	-0.2267	-0.2246	-0.2334	-0.2300
Road type: slip road	0.5335	0.5370	0.5123	0.5223
Speed limit 2	0.1730	0.1729	0.1830	0.1696
Speed limit 3	-0.0089	-0.0091	0.0010	-0.0043
Speed limit 4	-0.2127	-0.2108	-0.2131	-0.2159
Speed limit 5	-0.3658	-0.3652	-0.3590	-0.3534
Speed limit 6	-0.2205	-0.2189	-0.2222	-0.2260

RWMH output for the reported analysis. The CMC posterior summaries have smaller effective sample sizes than the corresponding full-data summaries in the computational output. These ESS values are not interpreted in isolation; they should be assessed together with trace plots, posterior plots, and other diagnostics.

No conclusion about computational efficiency is made because runtime values are not available in the computational output.

4.5 Diagnostic Plots

The computational output contains numerical posterior summaries and acceptance rates. However, trace plot files and posterior histogram or density plot files are not available.

TODO: insert trace plots for the full-data and Consensus Monte Carlo chains after the figure files are available.

Figure 4.1: Placeholder for trace plots

TODO: insert posterior histograms or density plots after the figure files are available.

Figure 4.2: Placeholder for posterior histograms or density plots

Additional convergence diagnostics such as autocorrelation plots or potential scale reduction factors are not available in the computational output.

TODO: compute or export additional convergence diagnostics if required.

4.6 Summary of Findings

Based on the computational output, the full-data Random Walk Metropolis–Hastings and full-data Independent Metropolis–Hastings samplers produce very similar posterior means, posterior standard deviations, and 95% credible intervals. This agreement provides a useful baseline for comparing the Consensus Monte Carlo results.

For Consensus Monte Carlo using Random Walk Metropolis–Hastings, the posterior summaries are generally close to the full-data posterior summaries. The CMC-RWMH subset acceptance rates are also close to the full-data RWMH acceptance rate. For Consensus Monte Carlo using Independent Metropolis–Hastings, posterior summaries are available, but the subset acceptance rates are lower and more variable than those from CMC-RWMH.

The results support a cautious descriptive conclusion that the Consensus Monte Carlo posterior summaries, especially those based on Random Walk Metropolis–Hastings, are broadly similar to the full-data posterior summaries for the computational output. However, no conclusion about computational efficiency is made because runtime values are not available. Further graphical diagnostics should be added after the trace plots and posterior histograms or density plots are exported.

CHAPTER 5
CONCLUSION

REFERENCES

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